

Failure of the 2024 Lost-Lepton Transfer-Factor Closure

Prefit validation report and analysis-method decision

The current MC-derived LLCR-to-zero-lepton transfer-factor method fails its data validation and must be retired from the nominal lost-lepton background estimation.

Executive summary

The combined transfer-factor estimate and a dedicated Top/W-split implementation were tested in independent MC folds and orthogonal data validation regions. The software and normalization chain closes in MC, but the method does not transfer from the one-lepton control region to data in the zero-lepton target.

- High-dM data residual closure reaches only 65-68% of the lost-lepton yield required by data.
- Top/W separation changes the high-dM result by less than 0.4 percentage points.
- In two low-dM validation regions, raw MC is within 1-3% of the data residual, while TF application degrades the agreement to 78-90%.
- A held-out low-dM Nb=1 control category is underpredicted by 21% with a pull of -14.48.
- The failed method is archived for diagnosis only and must not define nominal datacard rates.

Scope: 1,153 ROOT files, 361,054,245 scanned events, and 34,735,958 selected events. The selection authority is `real_subset_worker.py`. Nominal intermediates were not modified. Statistical covariance is included; detector and modeling systematics are not.

CMS Work in progress - 2024 data at 13.6 TeV

1. Method and validation strategy

The combined method multiplies the background-subtracted one-lepton data yield by an MC ratio of zero-lepton to one-lepton target-process yields. The closure observable is the predicted lost-lepton yield divided by the zero-lepton data residual after subtraction of Z to neutrinos, DY, photon+jets, diboson, and QCD multijet.

Combined TF: prediction = TF(combined TTbar + W+jets + single-top) x (one-lepton data - other MC).
Top/W split: prediction = mu(Top) x Top zero-lepton MC + mu(W) x W+jets zero-lepton MC.

Top/W normalization fit

Regime	Top normalization	W+jets normalization	Correlation
High-dM	0.8823 +/- 0.0101	1.0504 +/- 0.0175	-0.659
Low-dM	0.9514 +/- 0.0178	0.7300 +/- 0.0086	-0.083

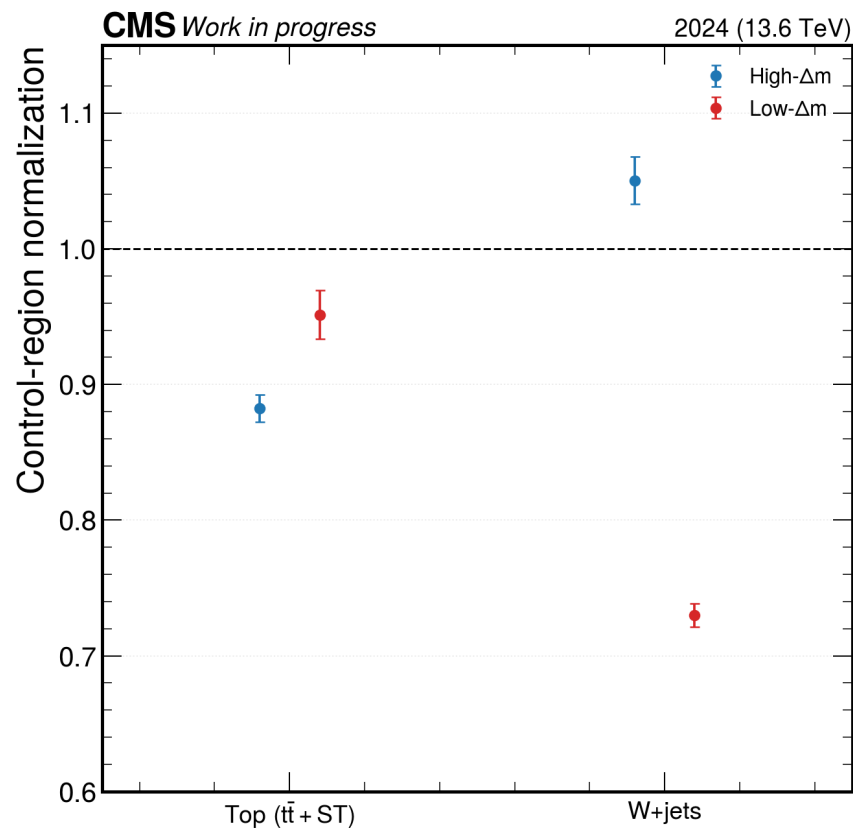


Figure 1. Independent Top and W+jets normalizations extracted from enriched one-lepton control categories. The uncertainties shown are statistical.

2. Technical closure and control diagnostics

The independent A/B-fold MC closure has maximum pulls below 0.21, and the full-mixture MC pseudodata closure has maximum pulls below 0.19. These tests validate the histogram reduction, normalization bookkeeping, bin mapping, fold independence, and background subtraction. They cannot validate the generator-to-data transfer.

MC categorization	Valid bins	Chi2 / ndf	Max. pull
High-dM pTmiss	6 / 7	0.0029 / 6	0.054
High-dM search bins	29 / 60	0.0926 / 29	0.173
Low-dM search bins	36 / 42	0.1402 / 36	0.206

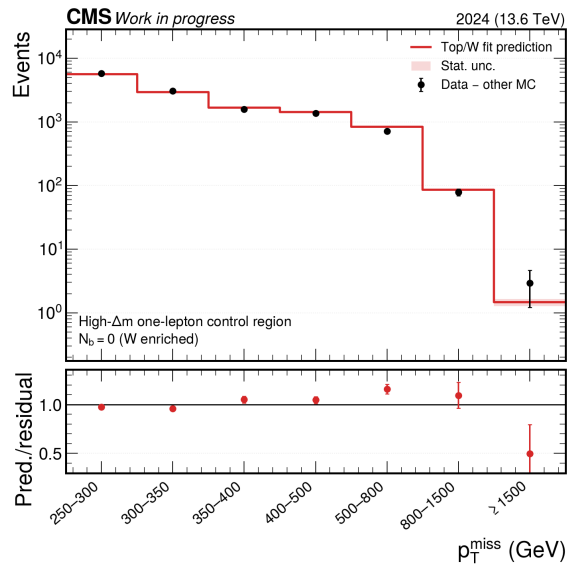


Figure 2. High-dM W-enriched one-lepton control shape. The integrated normalization is fitted, but the recoil shape remains incompatible ($p = 7.95\text{e-}4$).

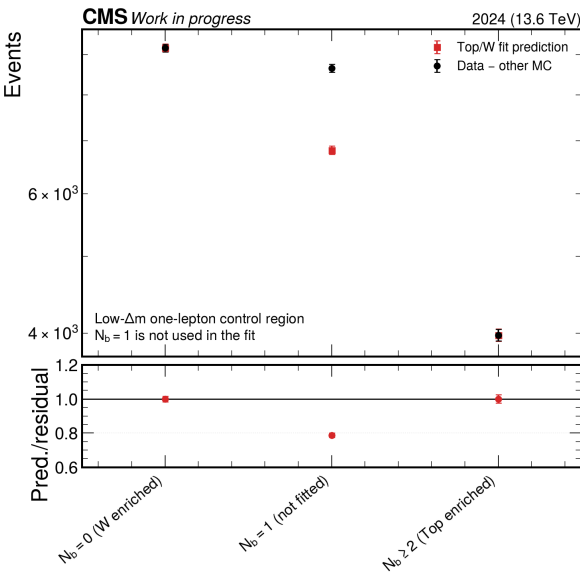


Figure 3. Low-dM N_b control validation. The held-out $N_b=1$ category is predicted at 0.788 of the data residual.

The low-dM $N_b=1$ category is not used in the normalization fit: prediction = 6814.8 +/- 81.2, data residual = 8649.9 +/- 97.3, and pull = -14.48.

3. Failure in high-dM data residual closure

The high-dM data residual requires substantially more lost-lepton yield than either TF implementation predicts. Top/W separation does not improve the result: the integrated ratios change from 0.6526 to 0.6539 in the Nb=0 region and from 0.6757 to 0.6794 in the Top-enriched region.

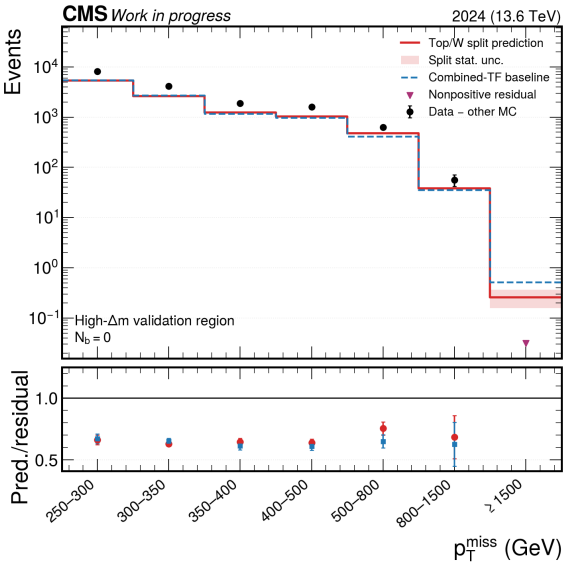


Figure 4. High-dM Nb=0 closure. The Top/W-split prediction reaches 0.654 of the data residual.

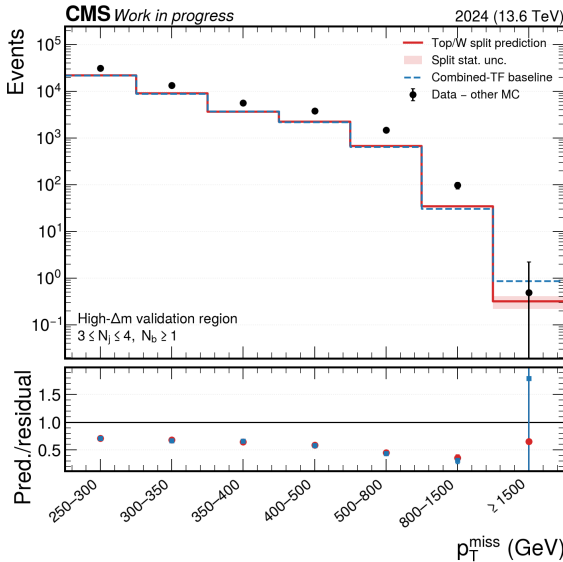


Figure 5. High-dM Top-enriched closure. The prediction reaches 0.679 of the data residual and develops a recoil-dependent deficit.

Region	Combined TF	Top/W split	p-value	Max. pull
High-dM Nb=0	0.6526	0.6539 +/- 0.0211	1.6e-75	14.51
High-dM Top enriched	0.6757	0.6794 +/- 0.0078	4.0e-268	19.64

4. Transfer-factor degradation in low-dM regions

The low-dM regions provide direct evidence that the LLCR-derived correction can be harmful. Raw lost-lepton MC is already close to the data residual, while the combined TF and the Top/W-split method move the prediction away from unity.

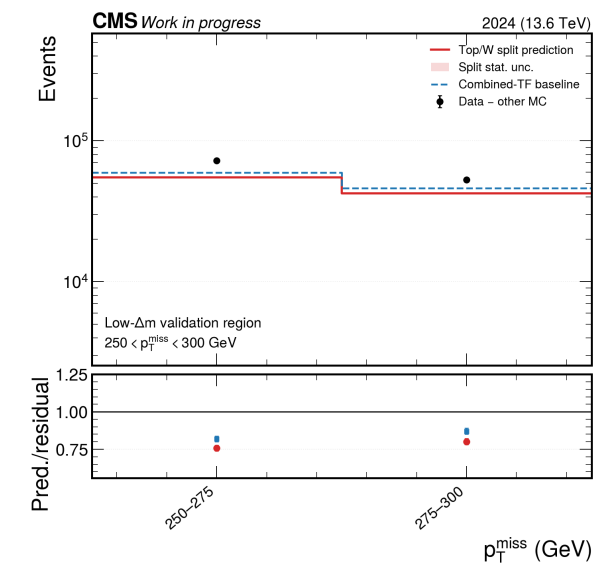


Figure 6. Low- $p_{T^{\text{miss}}}$ validation. Raw MC is at 0.990, while the combined and split TF predictions are 0.842 and 0.776.

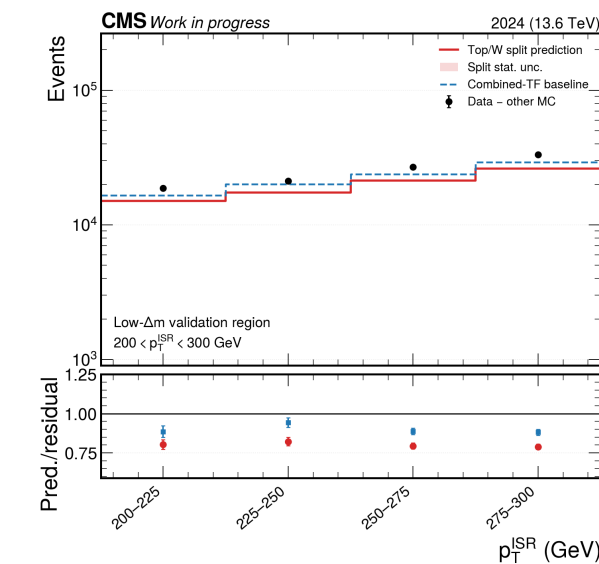


Figure 7. Low-ISR validation. Raw MC is at 1.025, while the combined and split TF predictions are 0.897 and 0.800.

Validation region	Raw LL MC	Combined TF	Top/W split
High-dM Nb=0	0.6438	0.6526	0.6539
High-dM Top enriched	0.7332	0.6757	0.6794
Low-dM low ISR	1.0254	0.8966	0.8001
Low-dM low $p_{T^{\text{miss}}}$	0.9904	0.8417	0.7763
Low-dM low MET significance	1.1326	1.0197	0.9911

Key observation: in low- $p_{T^{\text{miss}}}$ and low-ISR validation regions, the raw MC agrees at the 1-3% level. The TF methods import an LLCR-specific discrepancy and degrade the prediction by 10-22 percentage points.

5. Compatible sideband and why it does not rescue the method

The low-MET-significance validation region is compatible with closure after the Top/W split. It has only three valid bins, a large integrated uncertainty, and modest lost-lepton purity in the zero-lepton target. Its agreement cannot validate a method that fails in all other constraining validation regions.

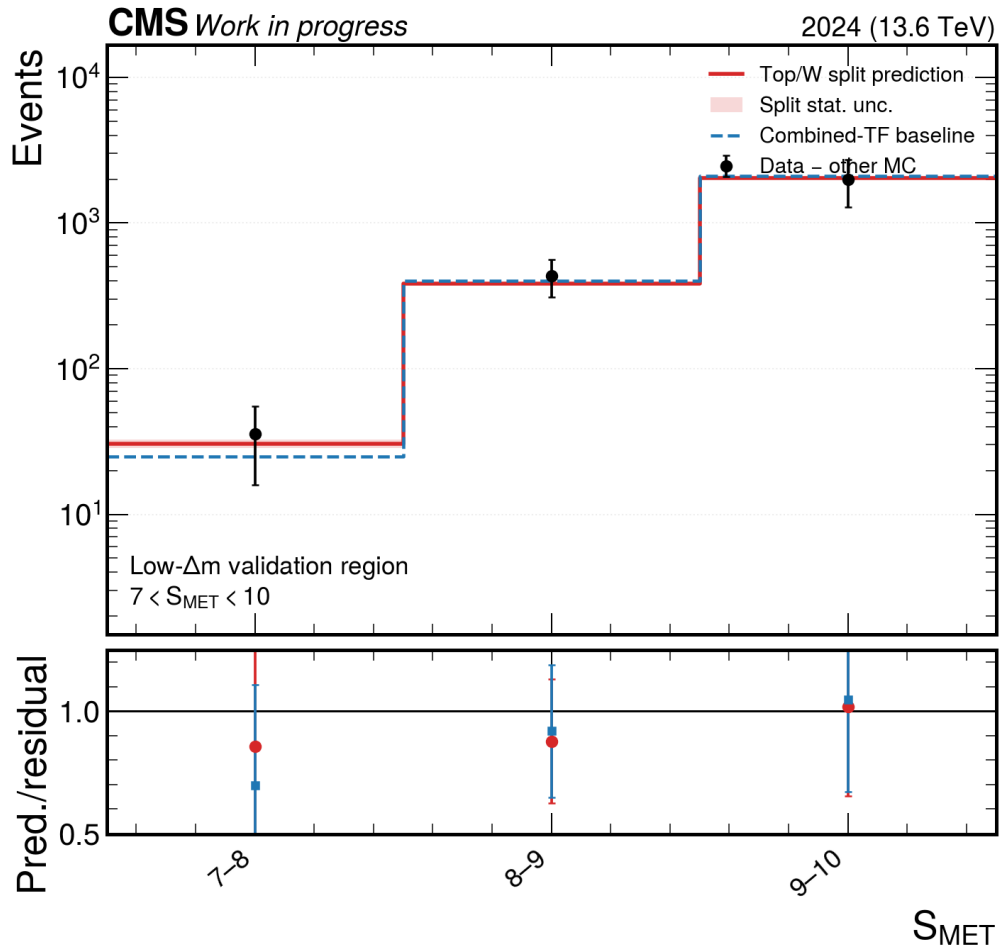


Figure 8. Low-MET-significance validation. The integrated Top/W-split ratio is 0.991 ± 0.292 . This statistically weak agreement is retained as a cross-check, not used to override the failed closure elsewhere.

Selecting only the sideband in which the correction happens to improve agreement would be an unjustified a posteriori choice. A valid transfer method must be portable across independently defined validation regions.

6. Analysis decision and consequences

Retire the current MC-derived LLCR transfer-factor method.

- Do not use either the combined or Top/W-split TF prediction as the nominal lost-lepton central value.
- Do not propagate the failed TF estimate into nominal datacards.
- Do not convert the observed 20-50% nonclosure into an empirical correction without an independently validated model.
- Preserve the code, JSON outputs, and figures as an archived failed-method study.
- Use raw lost-lepton MC only as a clearly labeled temporary reference. Raw MC does not solve the high-dM discrepancy.
- Adopt a replacement only after it improves closure relative to raw MC in held-out validation regions.

Recommended replacement

The preferred replacement is an event-level lepton-removal or lepton-embedding estimate. Electron and muon data events should be treated separately; the identified lepton momentum should be added to missing transverse momentum; recoil, angular selections, event categories, and search bins should be recomputed; and the event should be weighted by a data-measured probability for the lepton to fail the veto. Out-of-acceptance leptons and tau contributions must be explicit auxiliary components.

AN-ready conclusion

The MC-derived lost-lepton transfer-factor method was tested using independent MC folds and orthogonal data validation regions. Although the implementation closes in pure-MC and full-mixture MC pseudodata tests, it fails the data residual closure in both the high- and low-dM selections. In the high-dM validation regions, the method predicts only 65-68% of the lost-lepton residual observed in data. In two low-dM validation regions, the uncorrected lost-lepton MC agrees with the data residual at the 1-3% level, while application of the transfer factor degrades the agreement to 78-90%. Separating Top and W+jets does not restore closure and exposes additional inconsistencies in recoil shape and Nb migration. The transfer-factor method is therefore rejected as the nominal lost-lepton background-estimation strategy and is retained only as a documented failed validation study.

Supporting machine-readable results, source plots, and the complete Markdown report are stored next to this PDF.